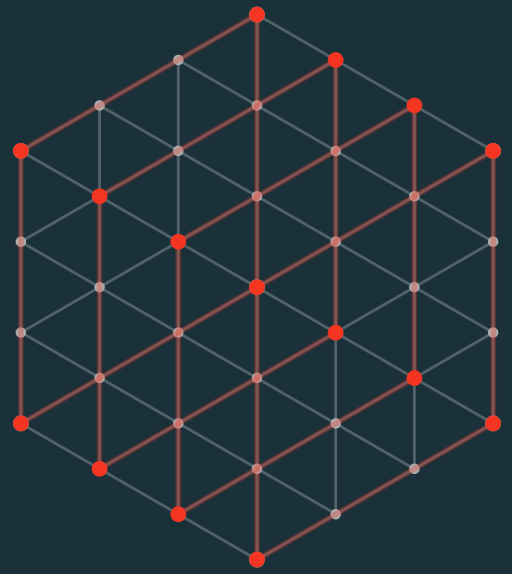
databricks

EASTWIND GOURMET · SPECIALTY FOOD RETAIL



Forecast-driven replenishment

You can’t forecast a gift box for a highway kiosk the way you do downtown.

For the VP Merchandising and the CIO

October 2026

THE OUTCOME

What you get

FORECAST ERROR

40–48%

lower than today's method, in every channel

WMAPE 0.24–0.25 vs 0.44–0.46 for last-year-same-week, across two holiday backtests

PLAN-FOR COVERAGE

91–92%

of store-SKU-weeks land within the p90 plan-for quantity

Calibrated p90 band drives safety stock, so you know how much buffer you are buying

LOST SALES RECOVERED · TARGET

$2–3M/yr

10–15% of today's $17.8M/yr lost sales

Conservative: the peak-season backtest cut modelled lost sales in every channel and every holiday window

VP Merchandising: fuller shelves on peak gifts, and orders your managers trust.

CIO: one governed flow from POS to order, traceable end to end.

THE PROBLEM

Ordering by gut feel

80 stores, 3 channels, one gift box that sells three ways

DOWNTOWN	HIGHWAY	AIRPORT
32	26	22
stores	kiosks	stores

Managers order from experience. Holiday peaks make the swings larger.



You lose both ways

$17.8M

lost sales a year from stockouts

$3.87M

expired and marked down a year

PEAKS AMPLIFY IT

18.7–21.1% of peak-gift store-days stocked out in Mid-Autumn 2025, vs 8.6% overall. Peak gifts carry 57.5% of lost sales.

FOR THE CIO

Why it’s hard

01

Five siloed sources

POS · ERP inventory · Product catalog · Store master · Holiday calendar

Different owners, formats, and refresh cycles. Even store IDs differ: POS says DT001, ERP says L0001.

02

Stockouts hide demand

When the shelf is empty, sales stop, so history understates what customers wanted.

One mooncake line was out of stock 6 and 7 days in its two peak weeks.

03

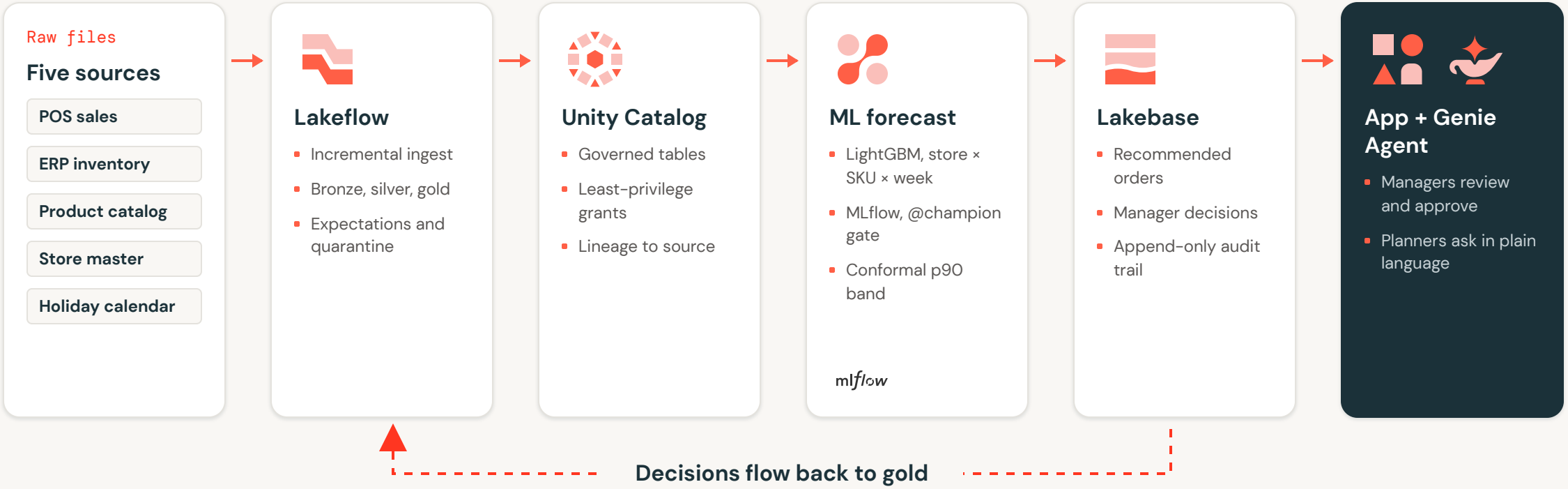
Festivals move

Lunar holidays shift week to week each year: Mid-Autumn fell on Sep 17 in 2024 and Oct 6 in 2025.

Last-year-same-week under-forecasts peak gifts by 30–39%.

ARCHITECTURE

One governed flow



One weekly job refreshes data, scores the @champion model, builds orders, loads Lakebase, and ingests decisions.

HOW IT WORKS

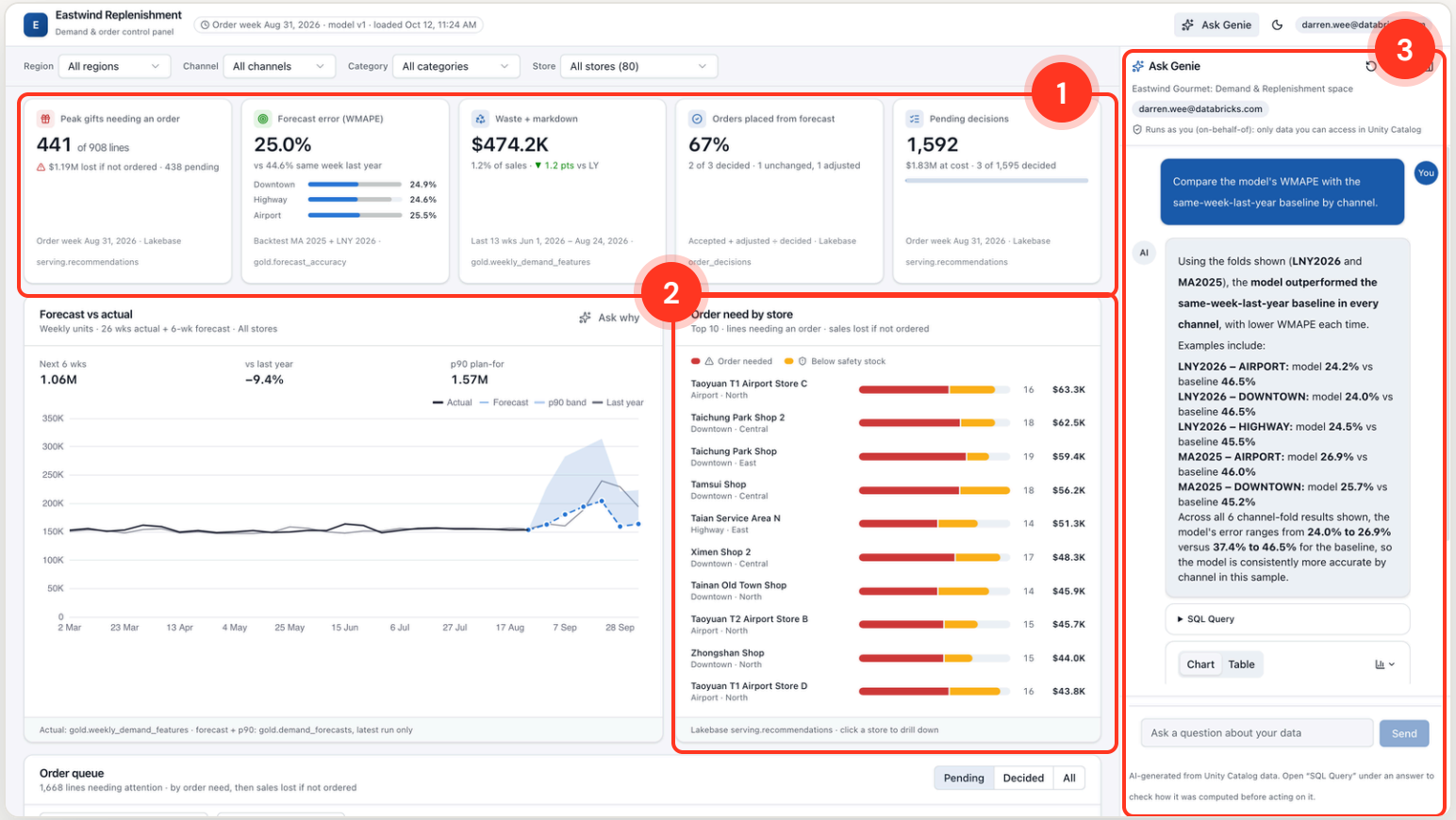
Pain point to result

PAIN POINT	DATABRICKS CAPABILITY	RESULT
Fragmented data	Lakeflow + Unity Catalog: one governed view	0 mismatched weeks; 40 bad rows quarantined
Hidden demand	Stockout-aware training excludes censored weeks	Peak-gift bias cut from 30–39% to 6–9%
Holiday swings	Festival-aligned features	Peak-gift error 0.25–0.31 vs 0.46–0.58
Distrust	Reasons in units, plus a p90 band on every line	Peak-gift demand within p90 in 90–95% of weeks
Slow decisions	One-click approve in the App, on Lakebase	Accept, adjust, or reject with a reason, all audited
Ad-hoc questions	Genie Agent on governed gold tables	11 of 11 benchmark questions correct, SQL shown

Error = WMAPE; baseline = last-year-same-week. Mid-Autumn 2025 and Lunar New Year 2026 backtests.

THE APP

What managers see



1 **KPI strip**
Peak gifts at risk, forecast error, waste, and adoption at a glance.

2 **Order queue**
Ranked by sales at risk. Accept in one click, or adjust with a reason.



3 **Ask Genie**
Plain-language questions, answered with your access and the SQL shown.

IMPACT AND PROOF

Proven vs next

SUCCESS MEASURE	EVIDENCE	STATUS	NEXT
Forecast accuracy by channel	WMAPE 23.6–26.6% vs 38.1–47.7% today	Proven	Track live by channel
Stockouts on peak gift SKUs	Modelled lost sales down in every window and channel Baseline: 18.7–21.1% of peak-gift days out	Simulated	Measure in the pilot
Expired and marked-down value	Flat: +$42k/yr in the backtest Baseline: $3.87M/yr	Not yet	Waste-aware policy, phase 2
Orders placed from forecast	Test decisions only so far	Not yet	Track in the pilot

PEAK-SEASON BACKTEST

$26.8M/yr

lost sales avoided vs manual ordering, pooled across 25 holiday weeks

Peak-window annualised, not a whole-year estimate. It shows the $2–3M/yr target is conservative.

Policy backtest is a simulation on synthetic data; a full-year replay comes before any finance claim.

ROADMAP

30 / 60 / 90 days

Day 30

Governance pack

Row filters by region, column masks on cost, persona groups, lineage in UC.

Day 60

Waste-aware ordering

Add shelf-life and waste limits to the order rule; replay a full year.

Day 90

10-store pilot

Managers order from the app; you measure all four success measures live.

THE ASK

Approve a one-region pilot

You get live proof on stockouts, waste, and adoption before a network-wide rollout.